

LAB EXPERIMENT RECORD

Department	AIDS
Course Code / Course Name	DSA0601 / Data Handling and Visualization
Experiment	Data Visualization using R
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SET 5 – WEBSITE ANALYTICS

Aim

To analyze and visualize product inventory data using bar charts, stacked bar charts, and scatter plots, and to develop an interactive dashboard for exploring product inventory information.

PROGRAM 1: LINE CHART – Daily Page Views

To create a line chart to visualize the daily page views of the website over time..

```
date <- c(
  "2023-01-01",
  "2023-01-02",
  "2023-01-03",
  "2023-01-04",
  "2023-01-05"
)
page_views <- c(1500, 1600, 1400, 1650, 1800)
plot(
  page_views,
  type = "o",
  xaxt = "n",
  xlab = "Date",
  ylab = "Page Views",
  main = "Daily Website Page Views",
```

```
col = "blue"

)

axis(

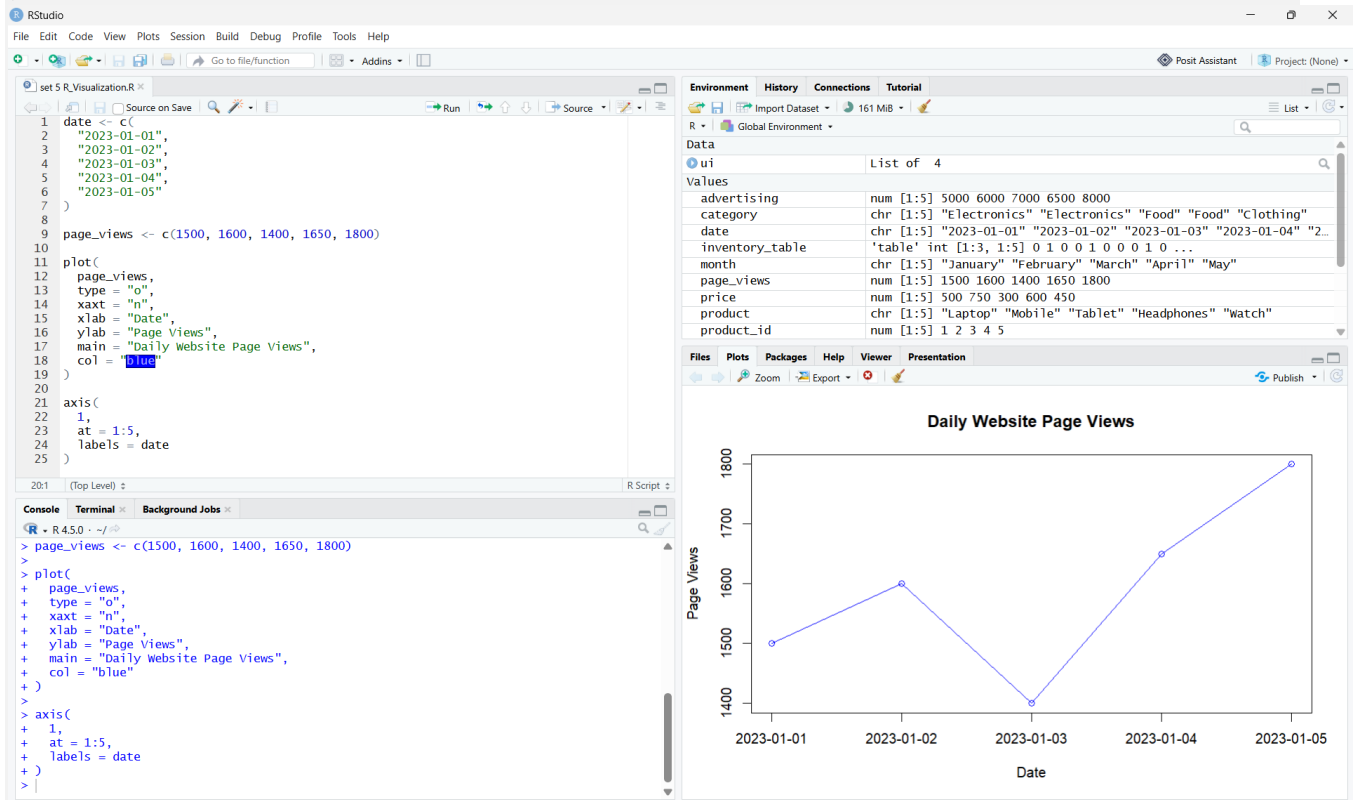
  1,

  at = 1:5,

  labels = date

)

)
```



Output 1: Daily website page views

Observation: The line chart was successfully created to visualize daily website page views. January 5 recorded the highest page views with 1800 views.

PROGRAM 2: BAR CHART

To create a bar chart to compare the click-through rate of the website across different days.

```
# date <- c(

  "2023-01-01",

  "2023-01-02",

  "2023-01-03",

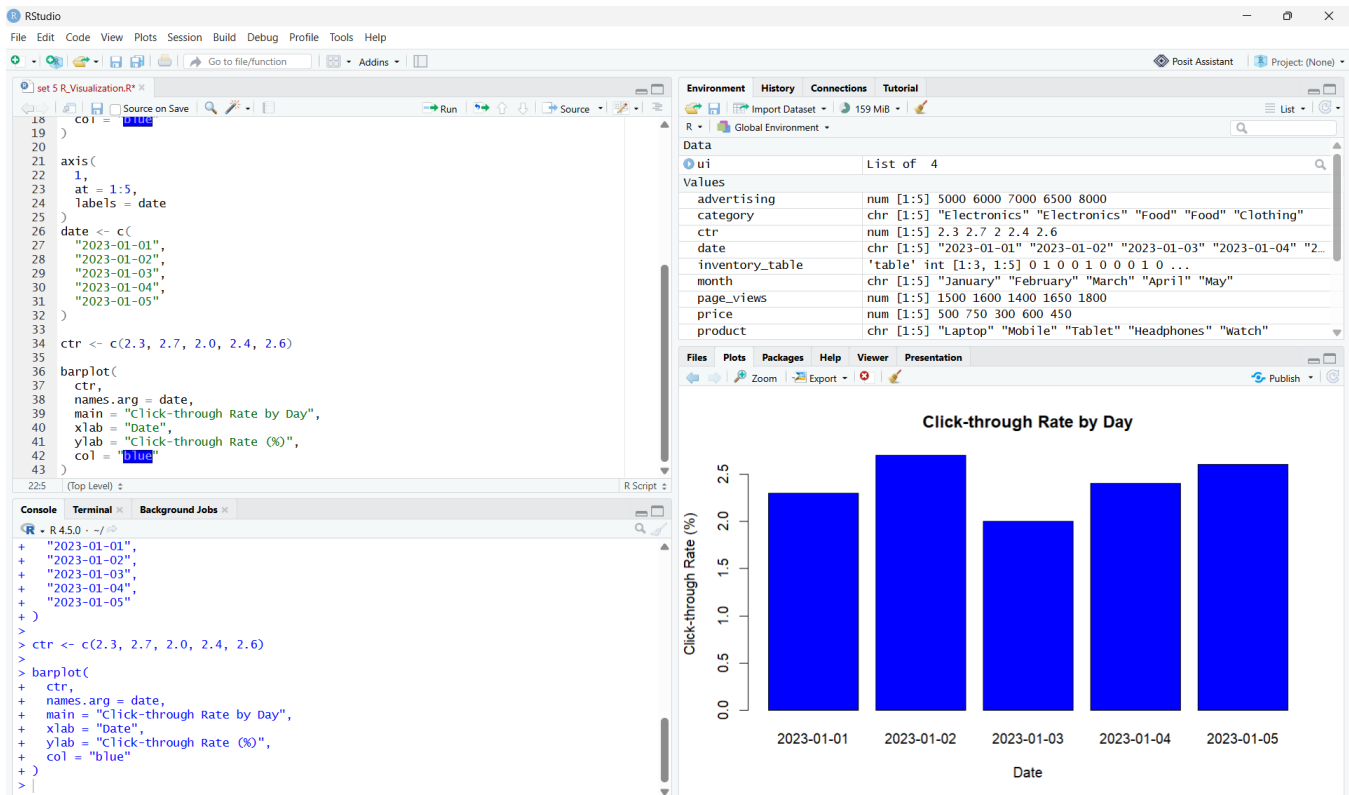
  "2023-01-04",

  "2023-01-05"

)
```

```
ctr <- c(2.3, 2.7, 2.0, 2.4, 2.6)

barplot(
  ctr,
  names.arg = date,
  main = "Click-through Rate by Day",
  xlab = "Date",
  ylab = "Click-through Rate (%)",
  col = "blue"
)
```



Output 2: click through rate by day

Observation: The bar chart was successfully created to compare the click-through rate for each day. January 2 recorded the highest CTR of 2.7%.

PROGRAM 3: Stacked Area chart – user interactions

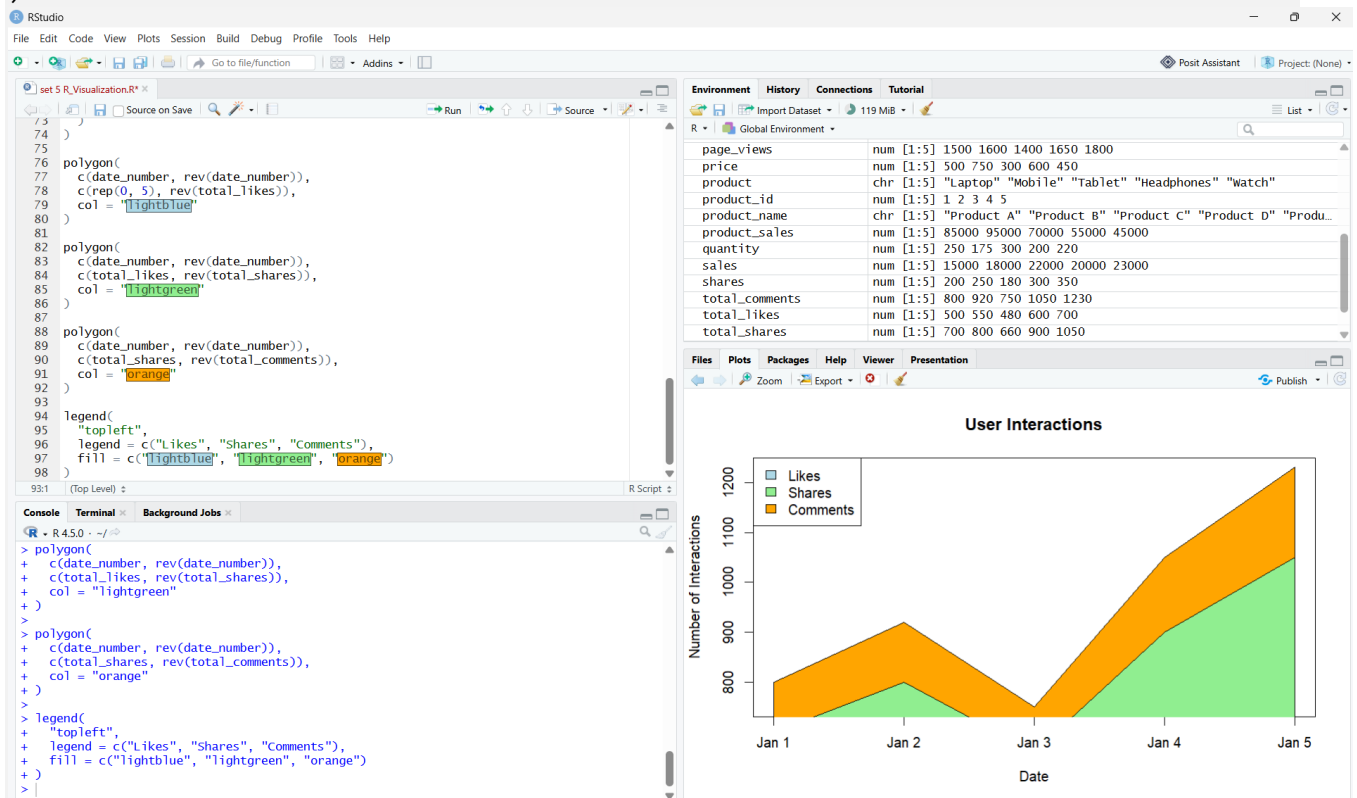
To create a stacked area chart to visualize different types of user interactions such as likes, shares, and comments over time.

```
# date_number <- 1:5
likes <- c(500, 550, 480, 600, 700)
shares <- c(200, 250, 180, 300, 350)
comments <- c(100, 120, 90, 150, 180)
total_likes <- likes
total_shares <- likes + shares
total_comments <- likes + shares + comments
plot(
  date_number,
  total_comments,
  type = "n",
  xaxt = "n",
  xlab = "Date",
  ylab = "Number of Interactions",
  main = "User Interactions"
)
axis(
  1,
  at = 1:5,
  labels = c(
    "Jan 1",
    "Jan 2",
    "Jan 3",
    "Jan 4",
    "Jan 5"
  )
)
polygon(
  c(date_number, rev(date_number)),
  c(rep(0, 5), rev(total_likes)),
  col = "lightblue"
)
```

```

polygon(
  c(date_number, rev(date_number)),
  c(total_likes, rev(total_shares)),
  col = "lightgreen"
)
polygon(
  c(date_number, rev(date_number)),
  c(total_shares, rev(total_comments)),
  col = "orange"
)
legend(
  "topleft",
  legend = c("Likes", "Shares", "Comments"),
  fill = c("lightblue", "lightgreen", "orange")
)

```



Output 3: user interactions

Observation: The stacked area chart was successfully created to visualize different types of user interactions over time.

PROGRAM 4: Interactive Dashboard

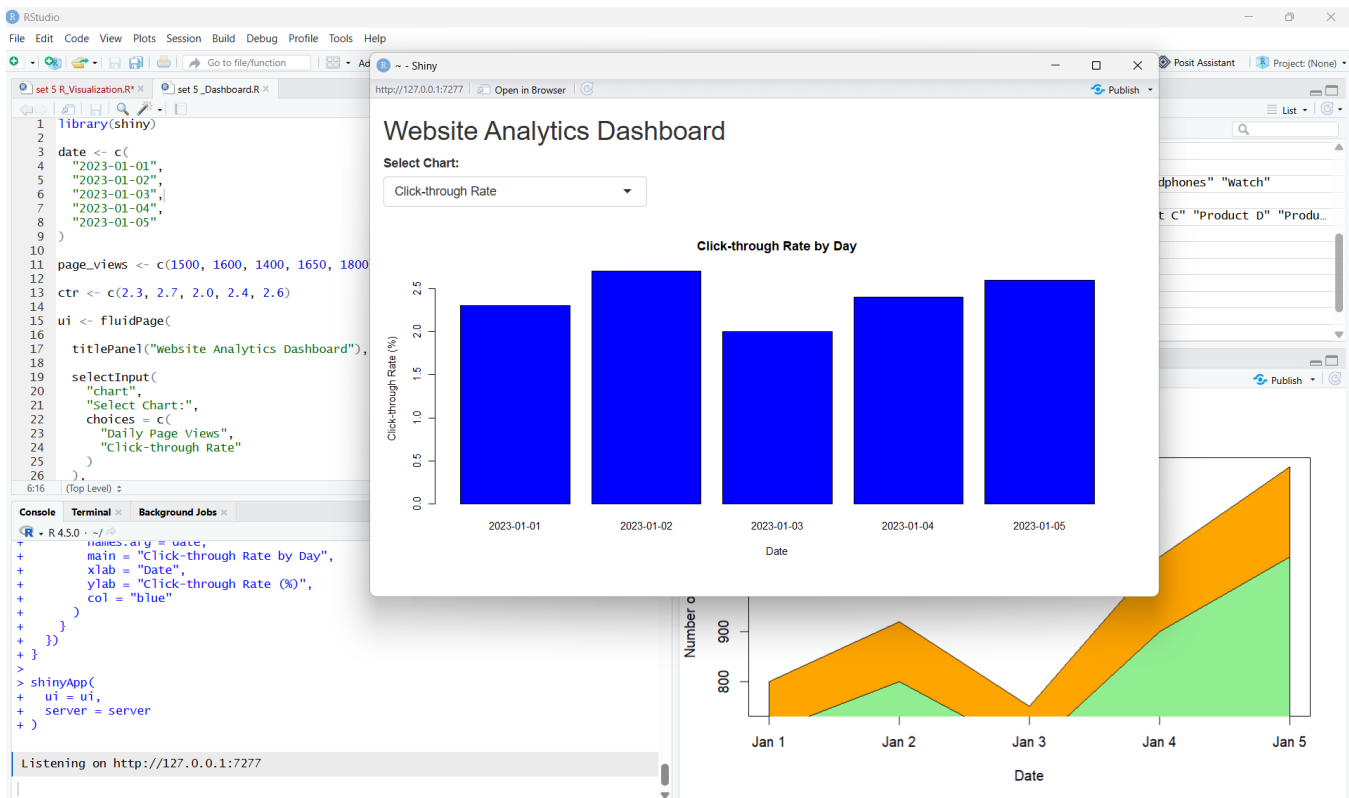
To develop an interactive dashboard combining the daily page-view line chart and click-through-rate bar chart.

```
library(shiny)
date <- c(
  "2023-01-01",
  "2023-01-02",
  "2023-01-03",
  "2023-01-04",
  "2023-01-05"
)
page_views <- c(1500, 1600, 1400, 1650, 1800)
ctr <- c(2.3, 2.7, 2.0, 2.4, 2.6)
ui <- fluidPage(
  titlePanel("Website Analytics Dashboard"),
  selectInput(
    "chart",
    "Select Chart:",
    choices = c(
      "Daily Page Views",
      "Click-through Rate"
    )
  ),
  plotOutput("myplot")
)
server <- function(input, output) {
  output$myplot <- renderPlot({
    if (input$chart == "Daily Page Views") {
      plot(
        page_views,
        type = "o",
        xaxt = "n",
        xlab = "Date",
        ylab = "Page Views",
        main = "Daily Website Page Views",
        col = "blue"
      )
    }
  })
}
```

```

    )
    axis(
      1,
      at = 1:5,
      labels = date
    )
  } else {
    barplot(
      ctr,
      names.arg = date,
      main = "Click-through Rate by Day",
      xlab = "Date",
      ylab = "Click-through Rate (%)",
      col = "blue"
    )
  }
})
}
shinyApp(
  ui = ui,
  server = server
)

```



Output 4: website analytics dashboard

Observation: The interactive website analytics dashboard was successfully developed. It allows users to switch between the daily page-view line chart and click-through-rate bar chart.

Set 4 Procedure

- Create vectors for the website dates, page views, and click-through rates.
- Use `plot()` to create a line chart showing daily page views over time.
- Use `barplot()` to compare the click-through rates for different days.
- Use sample values for likes, shares, and comments and create a stacked area chart using `polygon()`.
- Use `shiny()` to develop an interactive dashboard combining the daily page-view line chart and click-through-rate bar chart.
- Save each question as a separate R script and run the codes in RStudio to view the outputs.

Set 4 Result

The website analytics data was successfully analyzed and visualized using line, bar, and stacked area charts. The daily page views and click-through rates were compared, user interactions were visualized, and an interactive dashboard was successfully developed using RStudio and Shiny.